

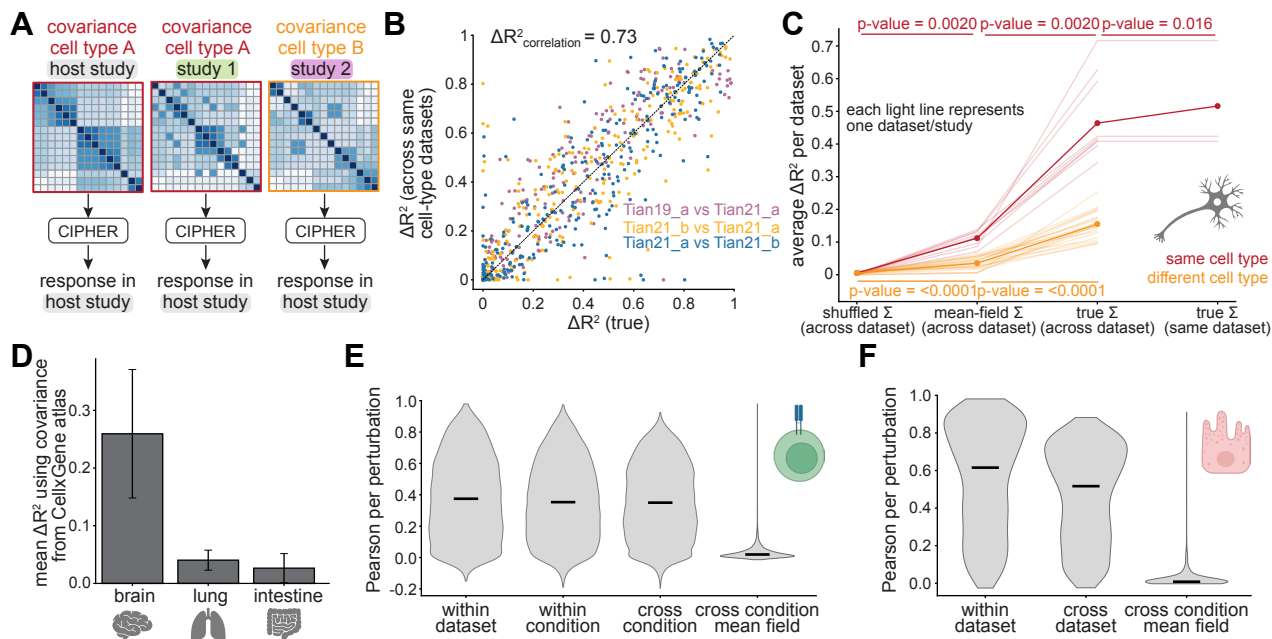


Fig. 4: CIPHER transfers across datasets and cell atlases.

a, Responses in a host study from inter-study covariances either of the same or different cell type. **b**, ΔR^2 values across neuron datasets. ΔR^2 values using the true covariance matrix from the host study compared to those from a different dataset but same cell type (ΔR^2 correlation = 0.73). Each point is a perturbed gene. **c**, Average ΔR^2 values across all pairs of datasets and conditions, using a covariance matrix from either a shuffled Σ (across datasets), the mean-field approximation (across datasets), the true covariance (true Σ , across datasets), or the true covariance from the same dataset. Thin lines correspond to individual dataset pairs (averages over all perturbations in the host dataset with a Σ of the same cell type in red or different cell types in orange) and points are averages over the typical ΔR^2 values for these pairs of datasets. Note that there are only three lines connecting the across and within dataset real covariance for the same cell type. This is because we are comparing the three datasets shown in **b**, and that there is no true covariance from the same dataset and different cell type. (Same cell type Shuffled vs. mean-field: Wilcoxon Signed-Rank Test, p-value = 0.0020; Same cell type mean-field vs. cross-dataset covariance: p-value = 0.0020; Same cell type cross-dataset covariance vs. same dataset covariance: p-value = 0.016. Different cell type Shuffled vs. mean-field: p-value < 0.0001; Different cell type mean-field vs. true: p-value < 0.0001). **d**, Mean improvement in prediction accuracy (ΔR^2) obtained by transferring covariance matrices derived from the CellxGene atlas to external perturbation datasets. Bars show the average increase in ΔR^2 relative to predictions using the corresponding mean-field covariance for brain, lung, and intestine datasets. Error bars denote variability across cell-types within each tissue. **e**, Distribution of Pearson correlations between predicted and real perturbation responses for T-cell perturbation datasets under different covariance transfer settings. Violin plots summarize prediction accuracy using covariance matrices estimated within the same dataset, within the same experimental condition, across different experimental conditions, and using a cross-condition mean-field covariance. Horizontal black bars indicate the mean Pearson correlation across perturbations. **f**, Distribution of Pearson correlations between predicted and real perturbation responses for RPE-1 perturbation datasets under different covariance transfer settings. Violin plots compare prediction accuracy using covariance matrices estimated within the same dataset, transferred across datasets (and labs) of the same cell-line, and using a cross-dataset mean-field covariance. Horizontal black bars indicate the mean Pearson correlation across perturbations.